B.Tech Major Project | Phase 1 Evaluation | Group 18

DEMENTIA DETECTION

Using EEG Signal Analysis & Deep Learning



Under the Guidance of Dr. Sujata Kulkarni

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Motivation

57.4M → 152.8M

Global Dementia Cases: 2019 to 2050

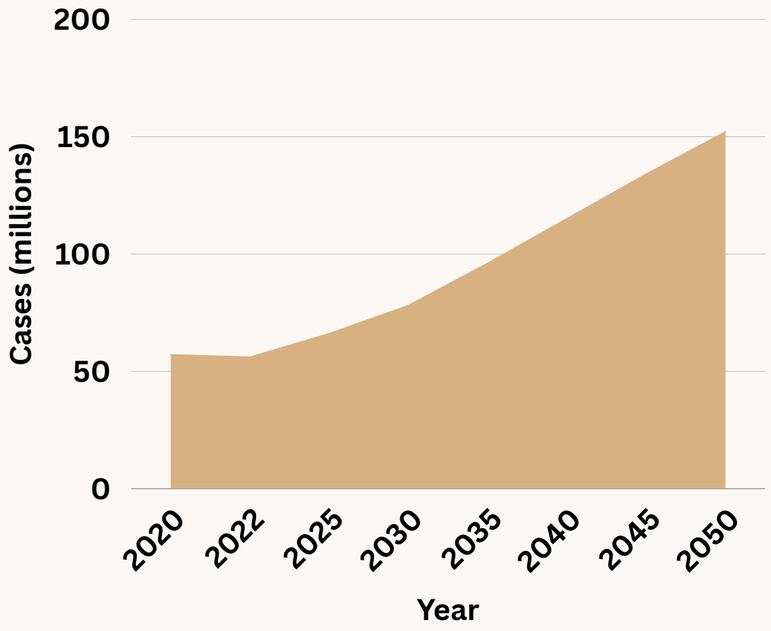
5.1M

cases in 2021

20M+

Projected by 2050

India specific



80%+

accuracy possible 9 years before symptoms

₹ 12 Lakh

saved per patient with early diagnosis

30%

reduction in institutional care

Why AI + EEG

Why CNN-Based EEG Analysis?

Non-Invasive — EEG captures brain electrical signals without surgery

Real-Time — Signals can be processed instantly unlike MRI

Cost-Effective — EEG equipment far cheaper than MRI scanners

AI-Ready — Deep learning excels at temporal pattern recognition

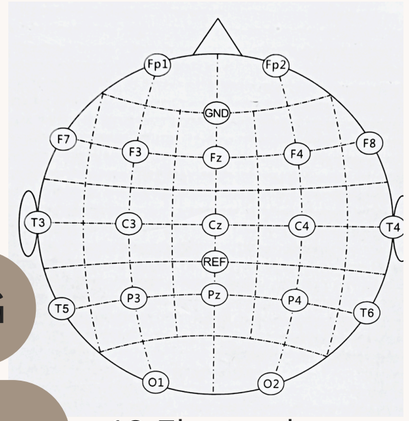
Research Objectives

- To explore and validate the authenticity of existing EEG dementia datasets
- To thoroughly analyze the CAUEEG and UCI EEG Alzheimer's datasets
- To use authentic research data to uncover hidden EEG patterns linked to dementia
- To increase number of clinically relevant features for dementia detection
- To implement existing algorithms to cross-validate the proposed model
- To develop an optimized deep learning model surpassing existing performance benchmarks

Key Datasets Used

PRIMARY: CAUEEG

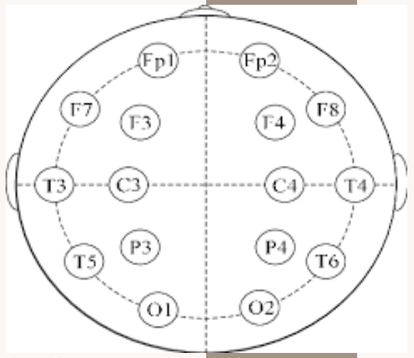
Authors: Kim, Youn, Paik et al., 2023



19 Electrode

- 19-channel resting-state EEG
- Clinically validated labels
- 3 classes: Normal, MCI, Dementia
- Includes eye open/closed annotations

CROSS-VALIDATION: UCI EEG AD



16 Electrode

- 848,640 records, 17 columns
- 16 float EEG features + 1 label
- No missing values
- Alzheimer's disease focused

Aspect	CEEDNet Paper	Our Project
Dataset	Only CAUEEG	CAUEEG + UCI EEG Alzheimer
Evaluation	Mostly within-dataset	Cross-dataset generalization
Focus	Model performance	Model robustness
Data Split	Allows some patient overlap	Strict patient-wise split

Literature Review

7 Papers Reviewed Across 3 Modality Categories

MRI Based Approaches

Alzheimer's Disease Neuroimaging Initiative (ADNI)

El-Assy et al.

2024 | 99.43%

Alzheimer's Disease Neuroimaging Initiative (ADNI) + OASIS

Odimayo et al.

2024 | 97-98%

ADNI-derived MRI dataset (6735 images)

Mousavi et al.

2025 | 98.52%

EEG Based Approaches

CAUEEG-based dementia classification using deep learning

Kim et al.

2023 | 79.16%

EEG dataset (AD, FTD, Healthy)

MDPI EEG Dataset

2023 | Clinical

Resting-state EEG

Jungrungrueang

2024 | 93.6%

Other Modalities

VOICE/AUDIO

Framingham Heart Study voice recordings

vkola-lab

Binary classif.

EEG-based approaches remain underexplored compared to MRI, and none achieve cross-dataset validation

Detailed review in next sldes

Research paper name	**Deep learning-based EEG analysis to classify normal, mild cognitive impairment, and dementia: Algorithms and dataset**
Link to paper	https://www.sciencedirect.com/science/article/pii/S1053811923002008?via%3Dihub
Author	Min-jae Kim, Young Chul Youn, Joonki Paik et al.
Dataset based upon	Chung-Ang University Hospital EEG (CAUEEG) A clinical EEG dataset with multi-class diagnostic labels including normal, mild cognitive impairment, and dementia.

Limitations

Limited Generalizability Outside CAUEEG
- Dataset is clinical and specific; performance on external populations is unknown.

Need for Large, Diverse EEG Data
- Deep models require many training samples to avoid overfitting.

Lack of Interpretability
- Deep learning models can lack explainability in clinical decisions.

Clinical Noise & Artefacts
- Real EEG signals may contain noise which model must learn to ignore

Methodology

- Dataset: Resting-state EEGs annotated for two tasks: multi-class dementia and binary (normal/abnormal) classification.
- Model (CEEDNet): An end-to-end deep learning framework that learns features directly from raw EEG, eliminating manual preprocessing.
- Evaluation: Benchmarked against traditional ML and CNN baselines using ROC-AUC and accuracy.

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Remediation

Transfer Learning Across Datasets
- generalization outside original demographic conditions.

Data Augmentation / Synthetic EEG
- Apply time-windowing, frequency distortions, and generative systems to expand effective dataset size without collecting new subjects.

Cross-Dataset Validation
- Evaluate the model on EEG datasets different from CAUEEG to test robustness and reveal dataset biases.

Benchmark

Model Architecture	#Params	Size (MiB)	TTA	Throughput (EEG/s)	Test Accuracy
CEEDNet (1D-VGG-19)	20.2M	77.2		7660.2	72.45%
CEEDNet (1D-VGG-19)	20.2M	77.2	✔️	998.54	74.28%
CEEDNet (1D-ResNet-18)	11.4M	43.5	✔️	844.65	74.85%
CEEDNet (1D-ResNet-50)	26.3M	100.7	✔️	837.66	76.37%
CEEDNet (1D-ResNeXt-50)	25.7M	98.2	✔️	800.49	77.32%
CEEDNet (2D-VGG-19)	20.2M	77.2	✔️	447.81	75.39%
CEEDNet (2D-ResNet-18)	11.5M	43.8	✔️	410.44	75.19%
CEEDNet (2D-ResNet-50)	25.7M	98.5	✔️	187.3	74.96%
CEEDNet (2D-ResNeXt-50)	25.9M	99.1	✔️	201.01	75.85%
CEEDNet (ViT-B-16)	86.9M	331.6	✔️	63.99	72.70%
CEEDNet (Ensemble)	253.8 M	969.9		26.4	**79.16%**

Research paper name	Alzheimer's Disease Neuroimaging Initiative (ADNI)	ADNI + OASIS	ADNI-derived MRI dataset (6735 images)
Link to Research paper	LINK	LINK	LINK
Author	A. M. El-Assy, Hanan M. Amer, H. M. Ibrahim, M. A. Mohamed	Simisola Odimayo et al.	Seyed Mohammad Mousavi, Khadijeh Moulaei, Leila Ahmadian
Methodology	• Dual CNN branches with different filter sizes • Concatenation before classification • Multi-class classification (3, 4, 5 categories) • End-to-end CN	• Structure-focused CNN • Gamma correction preprocessing • Midsagittal & parasagittal MRI planes	• Transfer learning • VGG16, VGG19, Xception, InceptionResNetV2 • Ensemble (majority voting)
Performace Indicator	99.43%, 99.57%, 99.13% accuracy (multi-class)	97–98% accuracy	High multi-class accuracy (reported >98%)
Positives	• Very high reported accuracy • Multi-class stage classification • Custom architecture (not just transfer learning)	• Focuses on structural atrophy regions • Domain-specific architecture	• Large dataset • Comparative architecture study • Ensemble framework
Problem Statement	Develop a CNN model capable of accurately classifying multiple AD stages from MRI scans.	Capture focal structural atrophy for improved AD classification.	Improve multi-class classification at scale.
Limitations	• Extremely high accuracy → possible overfitting • No strong cross-dataset validation • Limited explainability	• Only specific MRI planes • High computational complexity • Limited generalization testing	• Kaggle/derived dataset → may lack clinical diversity • Mostly 2D slice-based
Methods to fill the gap	• Use cross-validation + external validation dataset • Apply Grad-CAM for explainability • Apply regularization + dropout to avoid overfitting	• Use attention mechanisms to automatically learn focal regions • Compare 2D vs 3D inputs • Reduce model size for deployment	• Use 3D volumetric learning • Add clinical metadata fusion • Perform cross-site validation

Research paper name	EEG dataset (AD, FTD, Healthy)	Resting-state EEG	Framingham Heart Study voice recordings
Link to Research paper	LINK	LINK	LINK
Author	MDPI Data (EEG dataset creators)	Thawirasm Jungrungrueang et al.	vkola-lab
Methodology	Dataset publication (EEG recordings for dementia research)	• Dynamic EEG connectivity features • CNN-based classification	• CNN + LSTM on audio • Minimal feature engineering
Performace Indicator	It is not a model Paper its a clinicala dataset paper.	93.6% multiclass accuracy	Binary dementia classification
Positives	• Alternative modality (EEG instead of MRI) • Enables multimodal modeling	• Moves beyond static features • Uses dynamic connectivity	• Non-imaging modality • Real-world screening approach
Limitations	• No CNN benchmark model included • Dataset limited in size	• EEG only • No MRI integration	• No MRI comparison • Clinical data restricted
Problem Statement	Scarcity of public EEG dementia datasets.	Classify dementia subtypes using EEG biomarkers.	Automated voice-based dementia screening.
Methods to fill the gap	• Explore multimodal fusion (EEG + 1 Factor) • Customised CNN	• Combine dynamic EEG • Use spatiotemporal CNN	• Transformer based on Speech

Research Gap

Resource & Modality Gap

Over-reliance on expensive modalities (MRI) and massive architectures leaves affordable, lightweight EEG solutions underutilized.

Technical Gap

Existing EEG models fail to simultaneously analyze spatial and temporal features, missing key markers of dementia.

Validation Gap

A total lack of cross-dataset validation makes current model accuracies unproven and unreliable for real-world deployment

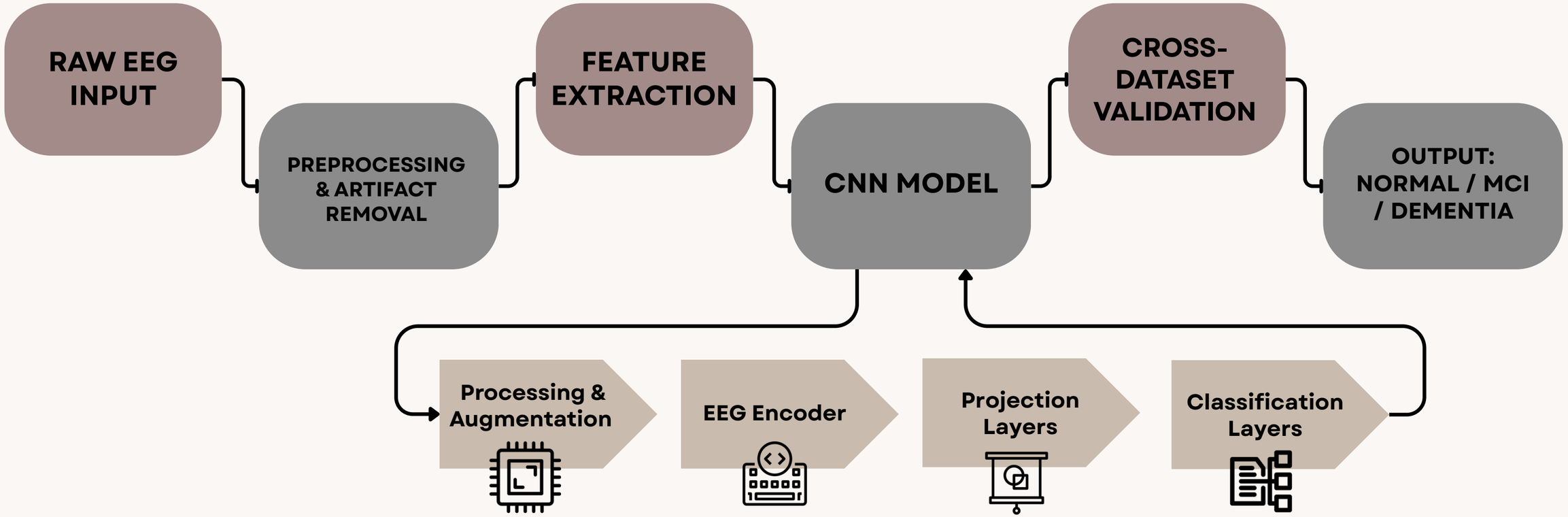
No existing work combines cross-dataset EEG validation with a lightweight spatiotemporal CNN for clinical deployment, THIS IS OUR OPPORTUNITY



Problem Statement

- Early dementia detection remains clinically underserved despite being the most critical window for effective intervention
- Existing approaches are either MRI-dependent (expensive, inaccessible) or use single-dataset EEG models with no cross-validation making reported accuracies unreliable for real-world clinical deployment.
- A lightweight spatiotemporal CNN trained on CAUEEG and rigorously cross-validated on the UCI EEG alzheimer's dataset delivering a clinically deployable, generalizable dementia detection model.

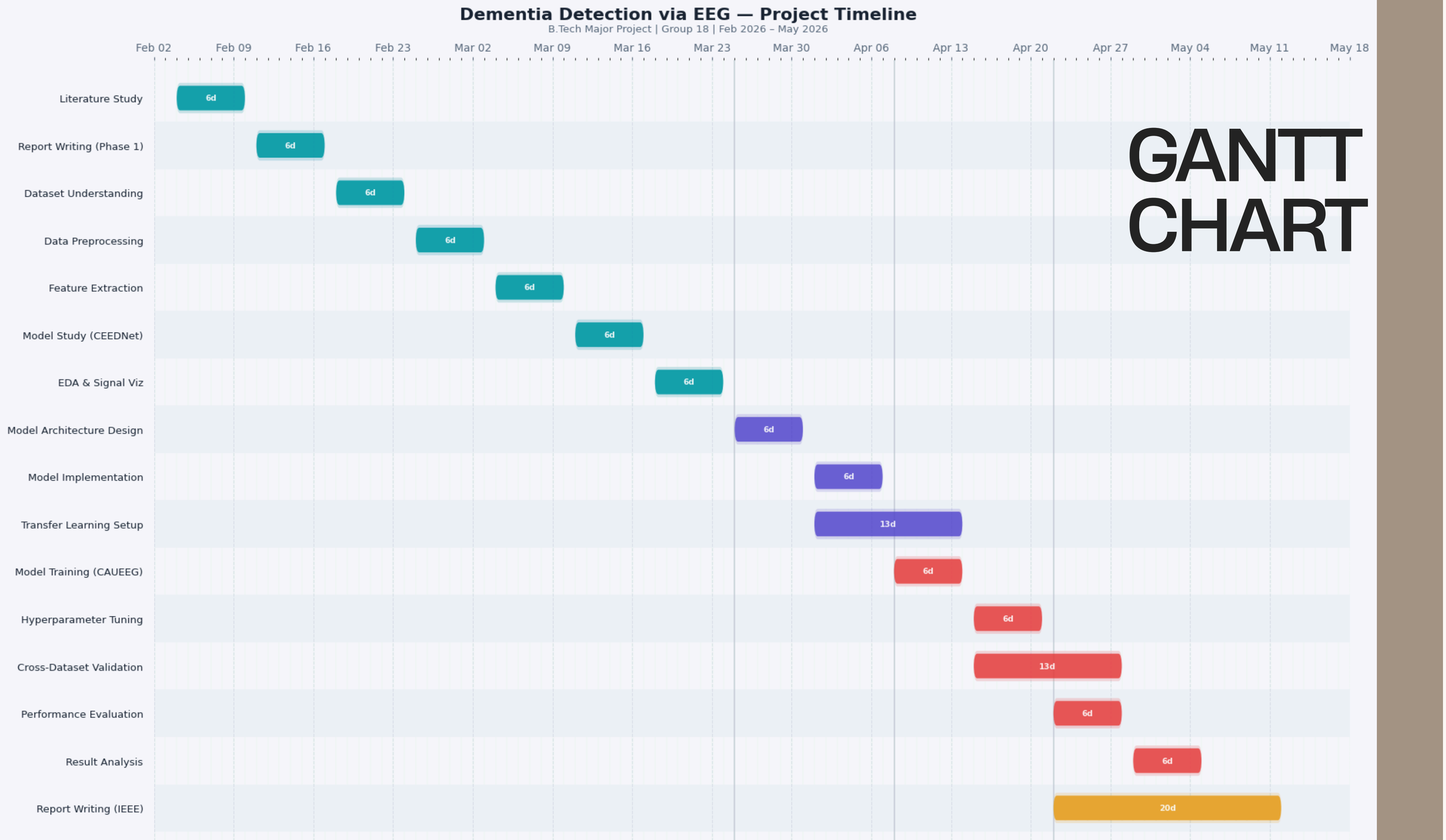
Proposed Framework



01 We evaluate a convolutional neural network architecture (e.g., 1D-ResNet-18 or 1D-VGG-19) specifically designed and benchmarked for the CAUEEG dataset.

02 We propose QSFE-Net: a custom four-stream architecture encoding frequency slowing, coherence loss, complexity reduction, and hemispheric asymmetry, integrated through a learned gating mechanism.

03 We validate QSFE-Net on the UCI EEG Alzheimer's dataset, a completely independent dataset, to test real-world generalisability, addressing the cross-dataset validation gap no existing EEG model has tackled.


Dementia Detection via EEG — Project Timeline
B.Tech Major Project | Group 18 | Feb 2026 – May 2026
Feb 02
Feb 09
Feb 16
Feb 23
Mar 02
Mar 09
Mar 16
Mar 23
Mar 30
Apr 06
Apr 13
Apr 20
Apr 27
May 04
May 11
May 18
Literature Study
Report Writing (Phase 1)
Dataset Understanding
Data Preprocessing
Feature Extraction
Model Study (CEEDNet)
EDA & Signal Viz
Model Architecture Design
Model Implementation
Transfer Learning Setup
Model Training (CAUEEG)
Hyperparameter Tuning
Cross-Dataset Validation
Performance Evaluation
Result Analysis
Report Writing (IEEE)
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6d
20d
GANTT
CHART

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